UNIVERSITY OF PERADENIYA
DEPARTMENT OF PHYSICS

END SEMESTER EXAMINATION - SEMESTER II - 2017/2018

PH 102 - General Physics II

Time allowed: ***Three hours***

Answer **Five** questions selecting **Two** from Section **A**, **Two** from Section **B** and **One** from Section **C**.

(Use of calculators is allowed)

Electron charge (e)	$= 1.60 \times 10^{-19}$ C
Rest mass of electron (m_o)	$= 9.11 \times 10^{-31}$ kg
Permittivity of free space (ε_o)	$= 8.85 \times 10^{-12}$ F m^{-1}
Permeability of free space (μ_o)	$= 4\pi \times 10^{-7}$ H m^{-1}
Planck constant (h)	$= 6.63 \times 10^{-34}$ J s
Speed of light in vacuum (c)	$= 3.00 \times 10^{8}$ m s^{-1}
Boltzmann constant (k)	$= 1.38 \times 10^{-23}$ J K^{-1}
Avogadro's number (N_A)	$= 6.02 \times 10^{23}$ mol^{-1}
Rydberg constant (R_∞)	$= 1.00 \times 10^{7}$ m^{-1}
Wien's constant (C)	$= 2.09 \times 10^{-3}$ m K

SECTION - A

1. (a) Write down the Gauss' law in words.

   (b) Suppose that an infinitely long straight insulating cylindrical cable of radius R has a uniform volume charge distribution of ρ.

   (i) Obtain an expression for the electric field intensity at distance r from the axis of the cylindrical cable.

   (ii) Sketch the variation of the electric field intensity $E(r)$ with the distance r.

   (iii) Also plot a rough sketch of potential $V(r)$ versus r.

   (c) A long cylindrical insulating cable of radius 1.00 cm has a uniform positive charge distribution of $\rho = 2.00 \times 10^{-3}$ C m^{-3}. Find the electric field intensity (E) and the potential (V) at distances

   (i) 0.50 cm and

   (ii) 2.00 cm from the axis.

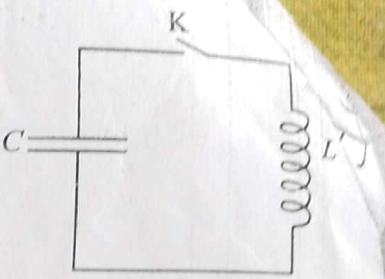
2. (i) Define the self-inductance and the mutual-inductance.

   (ii) Write down an expression for the magnetic flux density inside a thin hollow conducting cylinder of radius a carrying a steady current I.

   (iii) A coaxial cable consists of two thin metal conductors. The radius of the inner conductor is a and that of the outer conductor is b. The space between conductors are filled with a material having relative permeability μ_r. The conductors carry equal current I in opposite directions. Derive an expression for the self-inductance per unit length of the coaxial cable.

   (iv) If $a = 2.0$ mm, $b = 4.0$ mm and $\mu_r = 250$, find the self-inductance per unit length of the cable in part (iii).

3. A capacitor of capacitance C, an inductor of inductance L and a key K are connected in series as shown in the circuit diagram.



(a) Initially the capacitor is given a charge Q. The key K is then closed at time $t = 0$.

(i) Obtain an expression for charge q in the capacitor after time t.

(ii) Write down an expression for current I through the circuit at time t.

(iii) Draw the variation of current in the circuit as a function of time.

(iv) Write down an expression for the resonant frequency of the circuit.

(b) In the above circuit, the values of the capacitance $C = 2.0\ \mu F$ and the inductance $L = 80$ mH. Suppose that a potential difference of 5 V has been used to fully charge the capacitor.

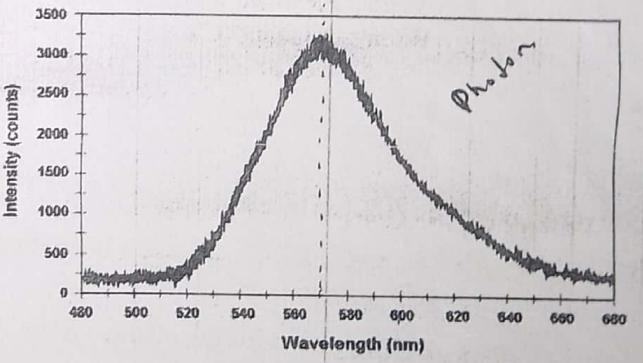
(i) Calculate the charge in the capacitor and the current in the circuit after time $t = 2.0$ ms.

(ii) Find the value of the resonant frequency of the circuit.

(iii) Name an electrical device that the above circuit is used.

SECTION B

4 (a) The figure shows the spectrum of light emitted by a firefly (an insect which emits light).



(i) Determine the temperature of a black body that would have a similar emission spectrum as shown in the figure.

(ii) Based on your result above, state whether firefly radiation is black-body radiation or not.

(iii) Name one application of the black-body radiation.

(b) In an experiment to study the photoelectric effect, a photoelectric cell with a metal target (cathode) having a work function of 1.95 eV and a photon source having an emission spectrum as in the above figure was used.

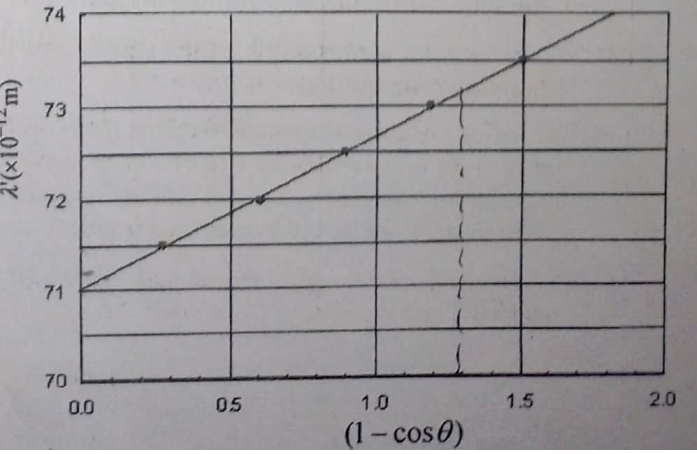
(i) Do you expect a photo-current to be produced? Justify your answer showing the necessary calculation.

(ii) Calculate the maximum kinetic energy of the photoelectrons emitted due to the peak radiation of the source.

(c) (i) Give two similarities and two differences between photoelectric effect and Compton effect.

(ii) Write down the equation for the Compton shift of the wavelength of a photon scattered by an electron. Clearly identify the symbols used.

(iii) Obtaining necessary information from the graph of λ' vs $(1-\cos\theta)$, calculate the kinetic energy of the recoiled electron when the incident photon scattered by 120°. Note $hc = 1240$ eV nm



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(i) Give a success and a failure of the Bohr model of atom.

(ii) State the Bohr postulate related to the transitions among electronic energy levels of an atom.

(iii) Calculate the shortest and the longest wavelengths of Lyman series of H-atom. To which region of the electromagnetic spectrum do these wavelengths belong?

b) (i) Draw a labeled schematic diagram of an X-ray tube.

(ii) How is the electron beam produced in the X-ray tube?

(iii) Name two types of metals that can be used as the target in X-ray tubes.

(iv) Draw a schematic diagram of an X-ray spectrum and explain the origin of the continuous and line spectrum.

(v) Draw the expected X-ray spectrum in the same diagram in part (b)(iv), if the accelerating potential is lowered.

(c) A metal target with the atomic number 29 and an accelerating potential of 50 kV was used to produce X-rays.

(i) Calculate the shortest wavelength of X-rays produced.

(ii) Determine the de Broglie's wavelength of the electrons in part (c) and compare it with your answer in part (c) (i).

(iii) Give one use of the Mosely's law.

Note $hc = 1240$ eV nm.

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(i) Define the binding energy (B) and the mean binding energy per nucleon ($\overline{B}$) of a nucleus.

(ii) Obtain an expression for the binding of a ${}^{A}_{Z}X$ nucleus.

(iii) Give a rough sketch of the variation of $\overline{B}$ with A.

(iv) Using the mean binding energy, briefly discuss the fission reactions of ${}^{235}_{92}U$ and the distribution of fission products.

(v) Calculate the mean binding energy per nucleon for ${}^{89}_{36}Kr$ nucleus.

You may use $m_n = 1.0087$ u, $m_p = 1.0078$ u, ${}^{89}_{36}Kr = 88.9176$ u and $u = 931.5$ MeV/c^2.

(vi) Find the value of x and y for the following nuclear reaction.

$${}^{235}_{92}U + n \rightarrow {}^{144}_{x}Ba + {}^{89}_{36}Kr + y\,n \ .$$

Where ${}^{235}_{92}U = 235.0439$ u and ${}^{144}_{x}Ba = 143.9227$ u

(vii) Calculate the total energy released (in J) by 1.00 kg of ${}^{235}_{92}U$ from the above nuclear reaction.

SECTION – C

7. (a) (i) Explain the crystalline and the amorphous states of a material.

(ii) Using the energy band-structure model, explain the electrical properties of solids.

(iii) Giving examples, briefly describe the hydrogen bonding and the three types of Van der Waals bonding.

(b) The potential energy for a system of two atoms can be represented by an expression of the form

$$U(r) = \frac{2.56 \times 10^{-6}}{r^9} - \frac{9.00}{r},$$

where r is the distance between the atoms in nm and the energy $U(r)$ is in eV.

(i) Identify the repulsive and the attractive potential energy terms.

(ii) Give the units of the value of 2.56×10^{-6} and 9.00 of the above equation in part (b).

(iii) Find the equilibrium distance between the two atoms.

(iv) Calculate the binding energy of the system in eV.

(v) Calculate the corresponding force constant of the system.

8. (a) (i) How can the rotational spectrum of a molecule be used to estimate the size of that molecule?

(ii) An HCl molecule is in its first excited state of its rotational energy levels. If the distance between its nuclei is 0.13 nm, what is the angular speed of the molecule about its center of mass? (atomic masses of H = 1.0 u and Cl = 35.0 u)

(b) (i) Bond strength of HCl is 428 kJ mol^{-1} and that of NaCl is 410 kJ mol^{-1}. Which molecule has the higher vibrational frequency? Justify your answer.

(ii) Qualitatively compare the vibrational spectra of the above two molecules.

(iii) The HCl molecule absorbs an IR photon with wavelength 3760 nm when transits from the ground state to the first excited state of its vibrational energy levels. Calculate the effective force constant of the bond between H and Cl.

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